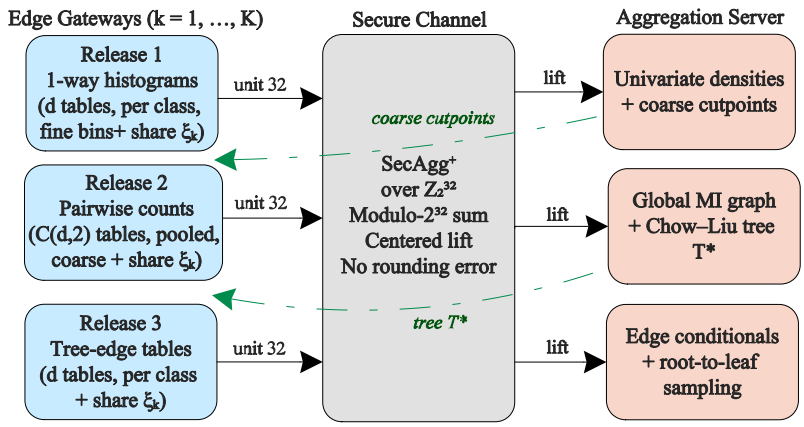

Edge Gateways (k = 1, …, K)
Secure Channel
Aggregation Server
Release 1
1-way histograms
(d tables, per class,
fine bins+ share ξ_k)
unit 32
Release 2
Pairwise counts
(C(d,2) tables, pooled,
coarse + share ξ_k)
unit 32
Release 3
Tree-edge tables
(d tables, per class
+ share ξ_k)
unit 32
coarse cutpoints
SecAgg+
over $Z_{2^{32}}$
Modulo-2^{32} sum
Centered lift
No rounding error
tree T*
lift
lift
lift
Univariate densities
+ coarse cutpoints
Global MI graph
+ Chow–Liu tree
T*
Edge conditionals
+ root-to-leaf
sampling